Most of the gram positive cell wall contain considerable amount of teichoic acid and teichuronic acid. In addition, they may contain polysaccharide molecules. The gram negative cell wall contains three components that lie outside the peptidoglycan layer. 1. Lipoprotein 2. Outer membrane 3.Lipopolysaccharide. Thus the different results in the gram stain are due to differences in the structure and composition of the cell wall (Figure 1.11). The difference between Gram Positive and Gram negative bacteria is given in Table 1.6.

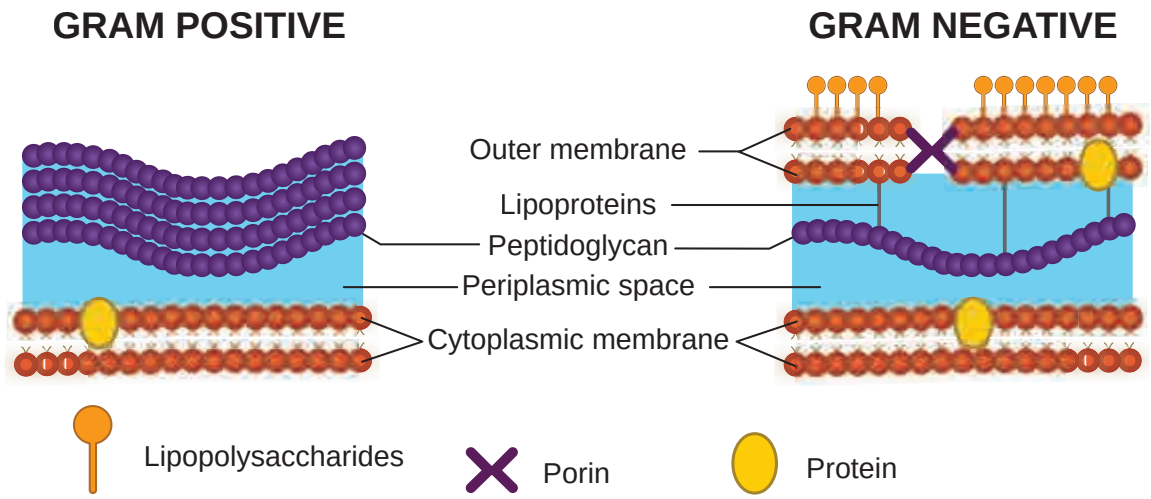


Figure 1.11: Difference between Gram positive and Gram negative bacteria

Table 1.6: Difference between Gram Positive and Gram Negative Bacteria

S.No	Characteristics	Gram positive Bacteria	Gram negative Bacteria
1.	Cell wall	Single layered with 0.015µm-0.02µm	Triple layered with 0.0075µm–0.012µm thick
2.	Rigidity of cell wall	Rigid due to presence of Peptidoglycans	Elastic due to presence of lipoprotein-polysaccharide mixture
3.	Chemical composition	Peptidoglycans-80% Polysaccharide-20% Teichoic acid present	Peptidoglycans-3 to 12% rest is polysaccharides and lipoproteins. Teichoic acid absent
4.	Outer membrane	Absent	Present
5.	Periplasmic space	Absent	Present
6.	Susceptibility to penicillin	Highly susceptible	Low susceptible
7.	Nutritional requirements	Relatively complex	Relatively simple
8.	Flagella	Contain 2 basal body rings	Contain 4 basal body rings
9.	Lipid and lipoproteins	Low	High
10.	Lipopolysaccharides	Absent	Present